الخوارزميات

الخوارزمية 1: المسار العام للدراسة

يربط هذا المسار مراحل الدراسة كلها، من البيانات إلى الاختبارات.

```
Algorithm 1: Overall study pipeline
Input : monthly data D (93 x 18), repetitions R = 10
Output: tables, tests, figures
1  EDA(D): descriptive statistics, ADF, IQR, VIF, PCA, bivariate relations
2  Remove accounting-identity variables (9)  -> features X (8), target y = TPF
3  for each scenario s in {IQR+random, no-removal+random, no-removal+chronological, 17 variabl
es}:
4      for r = 1..R:
5          (train, test) <- Split(D, s, seed r)                  # Algorithm 3
6          standardise with train mean / sd (target included)
7          V <- VAE(train); G <- cGAN(train)                      # Algorithms 5-6
8          for each model m in {LR,DT,RF,XGB,MLP,SVR,GBR,ET}:
9              for each training set T in {train, train U V, train U G}:
10                 fit m on T; predict test; store metrics        # Algorithm 7
11         fit Poisson/PPML on train; predict test                # Algorithm 11
12 Tests: Wilcoxon (augmentation), Friedman (models), DM (comparison)   # Algorithms 8-9
13 Interpret best hybrid: SHAP + learning curve                     # Algorithm 10
14 Hypothesis tests H1-H3                                           # Algorithm 12
```

الخوارزمية 2: كشف القيم الشاذة بقاعدة IQR

تُحذف كل مشاهدة تحتوي قيمة شاذة في أي متغير.

```
Algorithm 2: IQR outlier removal
Input : data D, columns J (target + explanatory variables)
Output: cleaned data D'
1  for each variable j in J:
2      Q1_j, Q3_j <- 25th and 75th percentiles of D[:, j]
3      IQR_j <- Q3_j - Q1_j
4      L_j <- Q1_j - 1.5 * IQR_j ;  U_j <- Q3_j + 1.5 * IQR_j
5      flag_ij <- ( x_ij < L_j ) or ( x_ij > U_j )
6  D' <- rows of D with no flag in any variable              # 93 -> 55 months
```

الخوارزمية 3: التقسيم والتوحيد القياسي

تُحسب معاملات التوحيد على التدريب وحدها لمنع تسرب الاختبار.

```
Algorithm 3: Split and standardisation
Input : data D', scenario s, seed r
Output: Z_train, Z_test
1  if s is random: permute rows with seed (1000 + r); train <- first 80 %, test <- rest
2  if s is chronological: train <- oldest 80 %, test <- newest 20 %
3  mu, sigma <- column mean and std (ddof = 0) of train, target included
4  Z_train <- (train - mu) / sigma ;  Z_test <- (test - mu) / sigma
```

الخوارزمية 4: انتقاء المتغيرات

```
Algorithm 4: Variable selection
Input : explanatory variables E, target y
1  for each e in E: r_e <- corr(e, y);  VIF_e <- 1 / (1 - R2_e) (regress e on the others)
2  Mark e as accounting-related if
       it is a component of y (sum of components ~ y), or
       it is a balance-sheet stock growing with y (|r_e| > 0.95 and VIF_e > 10)
3  Check: linear regression of y on all E  ->  R2 = 1.000 (leakage)
4  Keep X = E minus the marked variables ; recompute VIF (max 3.6)
```

الخوارزمية 5: المشفّر التبايني VAE

```
Algorithm 5: Variational Autoencoder (trained on train set only)
Input : Z_train (n x 9), latent dim 6, hidden 32, beta = 0.01, epochs 800
Output: n synthetic observations
1  Encoder: h = ReLU(W1 z);  mu = W2 h ;  log_var = W3 h
2  Decoder: z_hat = W5 ReLU(W4 u)
3  repeat 800 times (full batch, Adam lr 1e-3):
4      eps ~ N(0, I);  u = mu + exp(0.5 * log_var) * eps
5      L = mean((Z - decoder(u))^2) + beta * KL( N(mu, var) || N(0, I) )
6      update parameters by gradient descent on L
7  sample u ~ N(0, I) (n times) and return decoder(u)
```

الخوارزمية 6: الشبكة التوليدية الشرطية cGAN

```
Algorithm 6: Conditional GAN (trained on train set only)
Input : Z_train (n x 9), noise dim 8, 4 classes (quartiles of TPF), lambda = 10, epochs 4000
Output: n synthetic observations
1  c_i <- quartile class of the target of observation i (one-hot)
2  Generator G(noise, c): 64-64 ReLU layers -> 9 outputs
3  Discriminator D(z, c): 64-32 LeakyReLU layers -> 1 logit
4  repeat 4000 times (full batch, Adam lr 1e-4, beta1 = 0.5):
5      fake <- G(noise, c)
6      L_D = BCE(D(real, c), 1) + BCE(D(fake, c), 0)
7      L_G = BCE(D(fake, c), 1) + lambda * || mean(real) - mean(fake) ||^2   # feature matchin
g
8      update G on L_G, D on L_D
9  draw n classes with the same proportions as the training set; return G(noise, c)
```

الخوارزمية 7: التقييم الهجين المتكرر

```
Algorithm 7: Hybrid evaluation (one repetition)
Input : Z_train, Z_test, synthetic sets V and G
1  for m in {LR, DT, RF, XGB, MLP, SVR, GBR, ET}:        # hyper-parameters fixed (thesis Tabl
e 18)
2      for T in {Z_train, Z_train U V, Z_train U G}:
3          fit m on (T.X, T.y);  y_hat <- m(Z_test.X)
4          compute MAE, MSE, RMSE, R2, explained variance, median AE   (standardised units)
5          compute MAPE and RMSE in million MAD after back-transforming with (mu, sigma)
6  return the 24 rows of metrics for this repetition
```

الخوارزمية 8: اختبارا ويلكوكسون وفريدمان

```
Algorithm 8: Augmentation effect and model ranking
Input : RMSE[m, T, r] for r = 1..R
1  for each model m and generator g in {VAE, cGAN}:
2      a_r <- RMSE[m, train U g, r];  b_r <- RMSE[m, train, r]
3      p <- Wilcoxon signed-rank test of (a - b);  change% <- 100 * (mean(a) / mean(b) - 1)
4  Friedman: rank the 8 models (train only) within each repetition,
       test H0 "all models are equal" with chi-square statistic, report mean ranks
5  Count, for each model, the repetitions in which it has the lowest RMSE
```

الخوارزمية 9: اختبار ديبولد–ماريانو

```
Algorithm 9: Diebold-Mariano test (squared loss, Harvey correction, h = 1)
Input : forecast errors e1, e2 on the same test set (n obs)
1  d_t <- e1_t^2 - e2_t^2
2  DM <- mean(d) / sqrt( var(d) / n )
3  DM_H <- DM * sqrt( (n - 1) / n )
4  p <- 2 * ( 1 - CDF_t(|DM_H|, df = n - 1) )
5  repeat for every repetition; report the mean DM, the median p and the number of p < 0.05
```

الخوارزمية 10: SHAP ومنحنى التعلم

```
Algorithm 10: Interpretation of the best hybrid (Extra Trees + cGAN)
1  fit the model on (train U cGAN samples) of repetition 1
2  phi <- TreeSHAP(model, real train + real test)           # exact for tree ensembles
3  importance_j <- mean_i |phi_ij| ;  sort variables
4  learning curve: for fractions 0.2..1.0 of the augmented set,
       5-fold CV (shuffled), record mean train R2 and validation R2
```

الخوارزمية 11: نموذج Poisson/PPML المرجعي

```
Algorithm 11: Poisson / PPML benchmark
1  y_raw <- TPF in million MAD (train); X <- standardised explanatory variables
2  solve  sum_i [ y_i - exp(x_i' beta) ] x_i = 0   (GLM, Poisson family, log link)
3  predict exp(x' beta) on test; convert to standardised units; compute the same metrics
```

الخوارزمية 12: اختبار الفرضيات

```
Algorithm 12: Hypothesis tests H1-H3
H1: OLS of TPF on the 6 bank variables (levels, HAC errors, lag 3); same on growth rates
H2: add GR and IR; incremental F-test (levels and growth rates)
H3: for each model and 10 splits, RMSE(bank only) vs RMSE(bank + macro); paired
```

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